

DONGSHENG, China

WMO: 535430

Lat: 39.8155N

Lon: 110.0135E

Elev: 1459

StdP: 84.98

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.3	(c) -17.2	(d) -29.3	(e) 0.3	(f) -13.6	(g) -27.3	(h) 0.4	(i) -11.7	(j) 8.2	(k) -7.5	(l) 7.0	(m) -8.3	(n) 2.4	(o) 290	(p) 0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 29.9	(d) 15.6	(e) 28.6	(f) 15.3	(g) 27.3	(h) 15.0	(i) 19.7	(j) 24.7	(k) 18.8	(l) 23.7	(m) 18.0	(n) 23.1	(o) 3.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	15.7	21.7	17.3	14.8	21.5	16.1	13.7	20.8	63.5	24.7	60.1	23.8	57.1	23.1	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 6.5	(c) 5.7	(e) -22.0	(f) 32.2	(g) 2.5	(h) 1.7	(i) -23.8	(j) 33.5	(k) -25.3	(l) 34.5	(m) -26.7	(n) 35.4	(o) -28.5	(p) 36.7		
(a) 7.5	(b) 6.5	(c) 5.7	(e) -22.6	(f) 20.9	(g) 2.4	(h) 1.0	(i) -24.3	(j) 21.6	(k) -25.7	(l) 22.2	(m) -27.0	(n) 22.8	(o) -28.8	(p) 23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-8.8	-4.8	1.8	9.5	15.3	20.0	22.2	20.3	15.1	8.3	0.0	-6.9
	DBStd	11.51	4.53	5.05	5.39	5.13	4.09	3.11	2.63	2.77	3.63	4.33	4.77	4.48
	HDD10.0	2247	583	414	259	70	7	0	0	0	5	83	300	525
	HDD18.3	4165	842	648	513	267	109	19	2	13	107	312	550	783
	CDD10.0	1413	0	0	4	55	171	300	377	318	158	29	0	0
	CDD18.3	290	0	0	0	2	16	69	121	73	9	0	0	0
	CDH23.3	1928	0	0	0	18	123	488	864	386	48	0	0	0
	CDH26.7	413	0	0	0	2	14	99	228	61	8	0	0	0
Wind	WSAvg	2.8	2.5	2.7	3.1	3.3	3.1	2.9	2.7	2.6	2.6	2.6	2.8	2.7
Precipitation	PrecAvg	393	2	4	11	15	27	41	97	119	49	23	5	1
	PrecMax	710	13	25	45	84	108	121	190	243	131	52	27	5
	PrecMin	198	0	0	0	0	1	10	26	18	12	2	0	0
	PrecStd	117	3	5	10	18	22	29	52	62	29	15	7	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	9.9	18.7	25.6	28.5	31.0	32.5	29.9	27.8	21.4	13.7	5.0
		MCWB	-1.5	1.6	6.1	10.3	12.1	14.2	17.9	16.3	13.6	10.5	4.5	-1.0
	2%	DB	1.8	6.7	14.8	22.3	26.3	29.1	30.5	28.4	24.4	18.8	11.0	2.6
		MCWB	-3.4	-0.4	4.5	9.0	11.4	13.8	16.6	16.7	13.0	9.3	3.0	-2.5
	5%	DB	-0.6	4.4	12.2	19.9	24.6	27.7	29.1	26.9	22.7	16.8	8.8	0.7
		MCWB	-4.7	-1.4	3.1	7.8	10.8	13.4	16.2	16.2	12.5	8.1	2.2	-3.7
	10%	DB	-2.4	2.5	9.8	17.7	22.8	26.2	27.7	25.6	21.0	14.9	6.8	-1.0
		MCWB	-6.0	-2.6	1.8	6.6	9.8	13.3	16.1	15.8	12.2	7.3	1.4	-4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.2	2.1	7.4	12.6	14.3	17.9	21.1	20.9	17.5	12.5	6.2	-0.3
		MCDB	3.6	7.9	16.2	20.2	21.4	23.8	26.7	25.7	20.0	18.0	10.0	3.2
	2%	WB	-3.1	0.4	5.2	10.6	13.2	16.6	20.0	19.7	16.0	11.0	4.5	-2.1
		MCDB	1.6	5.0	13.4	18.5	21.2	22.3	25.3	24.1	20.5	15.8	8.8	1.7
	5%	WB	-4.6	-0.9	3.7	9.1	12.3	15.9	19.1	18.8	14.6	9.7	3.1	-3.3
		MCDB	-0.9	3.9	11.1	17.4	20.4	22.4	24.6	22.9	19.0	14.4	7.2	-0.1
	10%	WB	-5.9	-2.2	2.2	7.7	11.4	15.1	18.3	18.0	13.6	8.6	1.8	-4.3
		MCDB	-2.5	2.1	8.8	15.8	19.8	22.3	23.9	22.3	18.8	13.1	6.4	-1.1
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	9.5	10.1	10.1	9.6	8.8	8.3	8.4	8.6	8.0	7.5
		MCDBR	9.8	10.5	11.7	12.3	12.0	11.4	10.6	9.8	10.4	10.6	10.0	8.9
		MCWBR	6.8	6.4	6.4	5.9	5.1	3.9	3.3	3.4	4.0	5.0	5.5	6.1
	5% WB	MCDBR	9.4	9.5	11.1	10.7	10.3	9.3	8.2	7.8	7.9	8.9	8.5	8.0
		MCWBR	6.6	6.1	6.5	5.7	4.9	3.9	3.6	3.7	4.1	4.8	5.4	5.8
Clear-Sky Solar Irradiance	taub	0.312	0.370	0.403	0.451	0.481	0.531	0.502	0.506	0.473	0.412	0.344	0.316	
	taud	2.179	2.044	1.973	1.908	1.890	1.829	1.972	1.936	1.953	2.036	2.151	2.188	
	Ebn at Noon	834	823	838	823	807	765	784	767	761	765	783	794	
	Edh at Noon	103	137	164	187	195	208	179	180	166	137	105	95	
All-Sky Solar Radiation	RadAvg	2.71	3.61	4.99	6.08	6.72	6.42	6.08	5.49	4.65	3.89	2.91	2.38	
	RadStd	0.23	0.26	0.41	0.40	0.39	0.45	0.28	0.32	0.30	0.26	0.32	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.44	N/A	-1.42	N/A	N/A	N/A	N/A	-150	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	-1.14	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page